# Shahjalal University of Science & Technology

Department of Electrical and Electronic Engineering

$4^{th}$ year $1^{st}$ Semester Final Examination, 2021

Course Code: EEE 459

Course Title: Compound Semiconductor and Heterojunction Devices

**Total Marks: 100** **Credit: 3.0** **Time: 3 Hours**

***Answer any two questions from each part.***

## Part A

1. (a) Define mono crystalline, polycrystalline and amorphous materials. 3

   (b) What is effective mass? Explain a method with mathematical verification to determine effective mass from E-K diagram. 4

   (c) How the energy gap of $GaAs_{1-x}P_x$ is varied with mole fraction? Sketch the corresponding simplified E-k diagram. Discuss its light emission efficiency with and without iso-electric trap. 9

   (d) For an experimental setup an epitaxial layer of $In_xGa_{1-x}As_ySb_{1-y}$ is grown on InP substrate. If a light of wavelength less than 1243 nm passes through the epitaxial layer then it is absorbed in the film, otherwise it transmits through the layer. The lattice parameter of the unstrained InGaAsSb is 5.93 Å. find x and y using the following table. Also determine the critical thickness for this system. 7+2

| Semiconductor | Lattice Constant (Å) | Bandgap (eV) |
|---|---|---|
| InAs | 6.06 | 0.38 |
| GaAs | 5.65 | 1.43 |
| InSb | 6.48 | 0.18 |
| GaSb | 6.09 | 0.7 |

2. (a) What are the assumptions made in Anderson's model? Determine $\Delta E_C$, $\Delta E_V$ and $V_{bi}$ for an ideal n-Ge/p-GaAs heterojunction. Given that: Electron affinity of Ge = 4.13 eV, Electron affinity of GaAs = 4.07 eV, $n_i$ = [unclear: $2.5\times10^{13}$] $cm^{-3}$ (for Ge), $N_d = 1.2\times10^{16}$ $cm^{-3}$, and $N_a = 1\times10^{16}$ $cm^{-3}$. 2+5

   (b) Sketch the energy-band diagram of an abrupt $Al_{0.3}Ga_{0.7}As/GaAs$ heterojunction for:

   (i) [unclear: $N^+$]-AlGaAs, intrinsic GaAs.
   (ii) $P^+$-AlGaAs, n-GaAs.

   Assume $E_g$ of AlAs = 2.1 eV and $\Delta E_C = \Delta E_V$. 4+4

   (c) Consider an ideal tungsten-to-n-type GaAs junction. Assume the GaAs is doped to a concentration of $N_d = 5\times10^{15}$ $cm^{-3}$. Determine the theoretical barrier height, the built-in potential barrier, and maximum electric field for the case of zero applied bias. ($N_c$ = [unclear: $4.7\times10^{17}$] $cm^{-3}$, [unclear: $\epsilon_r$] 13.1, $\phi_m = 4.55$, $\chi = 4.07$) 10

3. (a) Write the conditions necessary to form rectifying contacts in metal-semiconductor junctions. Draw the energy band diagrams of metal-semiconductor rectifying junctions under ideal conditions at thermal equilibrium. Draw diagrams for both n-type and p-type semiconductors. 3+3+3


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(b) Discuss the significance of critical thickness in epitaxial growth. Draw enegy [*sic*: should be energy] band structure of Si and GaAs and show optical transitions in such semiconductors. 3+3+3

(c) Sketch the energy-band diagram of a tunneling junction. Why is this an ohmic contact? 7

## Part B

4. (a) From the Ebers-Moll equivalent circuit describe how a bipolar transistor operates. 8

(b) What is bandgap narrowing? 3

(c) Show that "The breakdown voltage in the open-base configuration is smaller, by the factor $\sqrt[n]{\beta}$ than the actual avalanche junction breakdown voltage." 9

(d) Consider a silicon emitter at T = 300 K. Assume the emitter doping increases from $10^{18}$ to $10^{19}$ $cm^{-3}$. Determine the new value of $p'_{E0}$ and determine the ratio $p'_{E0}/p_{E0}$. 5

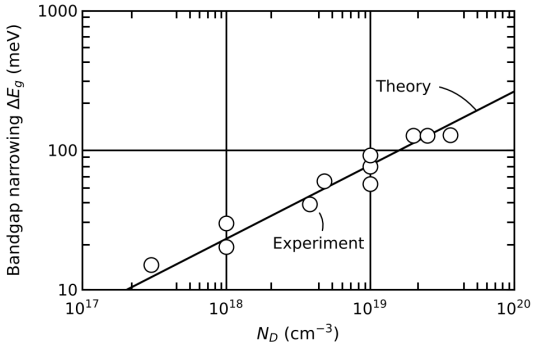



Figure 1: Fig. for Q. no. 4(d).

5. (a) With neat sketch, describe basic operation mechanism of TEBT. 4

(b) What is 2DEG? With quantum well structure describe the operation of HEMT. 5

(c) With the help of energy-band diagram and electric field profile, explain Kirk effect in BJT. How Kirk effect is minimized in HBT? Compare abrupt and graded junction HBTs in terms of performance parameters. 4+3+4

(d) Determine the two-dimensional electron concentration for an N AlGaAs – intrinsic GaAs heterojunction. Consider an $N - Al_{0.3}Ga_{0.7}As$ layer doped to $10^{18}$ $cm^{-3}$ and having a thickness of 500 Å. Assume an undoped spacer layer of 20 Å. Let $\phi_B = 0.85$ V and $\Delta E_c/q = 0.22$ V. The relative dielectric constant of $Al_{0.3}Ga_{0.7}As$ is 12.2. 5

6. (a) Draw the channel space charge region profiles of a single gate n-channel JFET for the following operating regions: (where $V_p$, $V_g$, and $V_d$ are pinch-off, gate, and drain voltages respectively)

(i) $V_g > V_p$ and $V_d = 0V$
(ii) $V_g > V_p$ and $V_d < V_{d,sat}$
(iii) $V_g > V_p$ and $V_d \approx V_{d,sat}$
(iv) $V_g > V_p$ and $V_d > V_{d,sat}$ 8

(b) In an n-channel MESFET, (where the symbols have their usual meaning) show that, $V_{po} = \dfrac{ea^2N_d}{2\epsilon_s}$ 6

(c) The $n^+p$ junction of a uniformly doped silicon p-channel JFET at T = 300 K has doping concentrations of $N_d = 10^{18}$ $cm^{-3}$ and $N_a = 10^{16}$ $cm^{-3}$. The metallurgical channel thickness is $a = 0.40$ $\mu m$. Determine the internal pinchoff voltage and the pinchoff voltage of the JFET. 5


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(d) Define transconductance for a JFET. Draw the small signal equivalent circuit of a JFET. 6

## Term Test 1 (its questions 1 and 2 are 7(a) and 7(b))

*[Course: EEE459, Term Test 1, Time: 30 Minutes, Full Marks: 20]*

7. (a) If $In_xGa_{1-x}As$ alloy is grown on GaAs substrate with critical thickness of 50Å, calculate the alloy composition. Lattice constants of InAs and GaAs is 6.058Å and 5.653Å respectively. 8

   (b) Why the effective mass of electrons in GaAs is different to that in Si? Which semiconductor has higher electron mobility? Discuss the variation of electron mobility in these semiconductors with electric field. 4+2+6


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